

Precept 4 - Solutions

Exercise 1 (Sum of Bernoulli Variables - Binomial, Geometric, Hypergeometric distributions)

ORF309 has $n + m$ students, of which n are girls and m are boys. During a lecture, students are randomly selected, one at a time, and given an action figure.

- (a) Assume a student may be selected multiple times, i.e. we *sample with replacement* and each student is equally likely to be selected. K action figures are given out. Let X be the number of action figures given to girls. Find $\mathbb{E}[X]$.
- (b) Assume a student may be selected multiple times, i.e. we *sample with replacement* and each student is equally likely to be selected. Let Y be the number of action figures given to girls before the first boy is selected. Find $\mathbb{E}[Y]$. Assume that the number of figures is unlimited ($K = \infty$).
- (c) Assume that each student may be selected only once, i.e. we *sample without replacement* and each of the remaining students is equally likely to be selected. Let Z be the number of action figures given to girls before the first boy is selected. Find $\mathbb{E}[Z]$. Assume that $K \geq n$.

Solution:

- (a) Each time an action figure is given out, it either goes to a girl or a boy. We define *success* as the outcome where an action figure is given to a girl and *failure* as one where it goes to a boy. The probability of success, p , is given by

$$p = \mathbb{P}\{\text{success}\} = \frac{n}{n+m}.$$

So if we think of giving out action figures as performing experiments, then each experiment is a *Bernoulli* variable. This is because each experiment results in success or failure, with a fixed probability of success (and thus a fixed probability of failure as well). Hence we can think of giving out K action figures as performing K independent experiments. It follows that X is a *binomial* random variable. Once this observation is made, the result easily follows by recalling that for $X \sim \text{Binom}(K, p)$, we have that $\mathbb{E}[X] = Kp = K \cdot \left(\frac{n}{n+m}\right)$.

- (b) Sticking with the notation defined above, let $p = \mathbb{P}\{\text{success}\} = \frac{n}{n+m}$, where success is, once again, defined as the event where an action figure is given to a girl. Let $q = 1 - p$. Clearly, q is the probability of failure (the event where an action figure is given to a boy), and it follows that $q = \mathbb{P}\{\text{failure}\} = \frac{m}{n+m}$. Similar to above, this is true because we are sampling with replacement and each student is equally likely to be selected.

If we think of giving out action figures as experiments, then Y is the number of experiments required until the first *failure*. From our lecture notes, we recognize this as a *geometric* random variable with parameter q . It follows that $Y \sim \text{Geom}(q)$. Note that here we are counting the number of successes and not the trials thus we need to subtract 1 from the expectation. Thus, we have $\mathbb{E}[Y] = \frac{1}{q} - 1 = \frac{n+m}{m} - 1 = \frac{n}{m}$.

(c) Define

$$Z_i = \begin{cases} 1 & \text{if girl } i \text{ is selected before any boy is selected} \\ 0 & \text{else} \end{cases}, \quad i \in \{1, \dots, n\}.$$

So

$$Z = \sum_{i=1}^n Z_i.$$

Hence,

$$\mathbb{E}[Z] = \sum_{i=1}^n \mathbb{E}[Z_i] = \sum_{i=1}^n \mathbb{P}\{Z_i = 1\}.$$

Let $i \in \{1, \dots, n\}$ and let us compute $\mathbb{P}\{Z_i = 1\}$.

There are three possible ways to compute this quantity.

First way: Note that the sample space of this random experiment is the set of all permutations of $n + m$ students, so we have $|\Omega| = (n + m)!$.

Each outcome (permutation) has probability $\frac{1}{(n+m)!}$ to occur, and we need to find how many different outcomes favor event $\{Z_i = 1\}$. First, pick $m + 1$ placements among $m + n$ in $\binom{m+n}{m+1}$ different ways. Then, we place girl i in the first position among the selected $m + 1$. The boys can be placed in the remaining m placements in $m!$ different ways. So, now we guarantee that girl i is picked before all the boys. We still need to place the remaining $n - 1$ girls to the $(m + n) - (m + 1)$ placements, and this can be done in $(n - 1)!$ different ways.

Thus, $\{Z_i = 1\}$ has

$$\binom{m+n}{m+1} m!(n-1)! = \frac{(m+n)!}{(m+1)!(n-1)!} m!(n-1)! = \frac{(m+n)!}{m+1}$$

many elements, and we have $\mathbb{P}\{Z_i = 1\} = \frac{1}{m+1}$. Then, it is easy to see that we have

$$\mathbb{E}[Z] = \frac{n}{m+1}.$$

Second way: Whether the other girls receive figures is irrelevant from the standpoint of the girl i . The only thing that matters is that she gets it before the boys. Thus, $P(Z_i = 1) = \frac{1}{m+1}$. Note that $Z = \sum_{i=1}^n Z_i$, and thereby $E[Z] = \sum_{i=1}^n E[Z_i] = \frac{n}{m+1}$.

Third way: Let us give a different argument for $\mathbb{P}\{Z_i = 1\} = \frac{1}{m+1}$. Denote by T^i the time at which girl i is selected. Note that if $Z_i = 1$, then $T^i \in \{1, \dots, n\}$. Hence,

$$\mathbb{P}\{Z_i = 1\} = \sum_{k=1}^n \mathbb{P}\{Z_i = 1, T^i = k\}.$$

Note that $\{Z_i = 1, T^i = k\}$ occurs if and only if we selected girls (other than girl i) in the first $k - 1$ times and we select girl i at time k . So, we have

$$\begin{aligned}\mathbb{P}\{Z_i = 1, T^i = k\} &= \frac{n-1}{n+m} \cdot \frac{n-2}{n+m-1} \cdots \frac{n-k+1}{n+m-k+2} \cdot \frac{1}{n+m-k+1} \\ &= \frac{(n-1)! \cdot (n+m-k)!}{(n-k)! \cdot (n+m)!} \\ &= \frac{1}{n} \cdot \frac{\binom{n+m-k}{m}}{\binom{n+m}{m}}.\end{aligned}$$

Therefore,

$$\begin{aligned}\mathbb{P}\{Z_i = 1\} &= \sum_{k=1}^n \mathbb{P}\{Z_i = 1, T^i = k\} \\ &= \frac{1}{n} \cdot \frac{1}{\binom{n+m}{m}} \sum_{k=1}^n \binom{n+m-k}{m} \\ &= \frac{1}{n} \cdot \frac{1}{\binom{n+m}{m}} \cdot \left(\sum_{k=0}^n \binom{n+m-k}{m} - \binom{n+m}{m} \right)\end{aligned}$$

To compute the sum in the above calculation, we use the recursive property

$$\binom{x}{y} + \binom{x}{y+1} = \binom{x+1}{y+1}, \quad x \geq y+1, \quad x, y \text{ positive integers},$$

of binomial coefficients. This property implies

$$\sum_{k=0}^n \binom{m+n-k}{m} = \sum_{k=0}^n \binom{m+k}{m} = \binom{m+k+1}{m+1}.$$

(Write the right-hand side as a sum of two coefficients and do this repeatedly to get a sum of $n+1$ coefficients.) Thus,

$$\mathbb{P}\{Z_i = 1\} = \frac{1}{n} \cdot \frac{1}{\binom{n+m}{m}} \cdot \left(\binom{m+k+1}{m+1} - \binom{n+m}{m} \right) = \frac{1}{m+1}.$$

Exercise 2 (Variance of a Poisson random variable)

Let $\lambda > 0$ and $X \sim \text{Poisson}(\lambda)$. Prove that $\text{Var}(X) = \lambda$.

Solution:

We know that $\text{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2$, and we know $\mathbb{E}(X) = \lambda$, thus all we need to calculate is the second moment:

$$\begin{aligned}\mathbb{E}(X^2) &= \sum_{k=0}^{\infty} k^2 \mathbb{P}(X = k) = \sum_{k=1}^{\infty} k^2 \frac{\lambda^k e^{-\lambda}}{k!} = \sum_{k=1}^{\infty} (k^2 - k) \frac{\lambda^k e^{-\lambda}}{k!} + \sum_{k=1}^{\infty} k \frac{\lambda^k e^{-\lambda}}{k!} \\ &= 0 + \sum_{k=2}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k-2)!} + \mathbb{E}(X) = \lambda^2 \sum_{k=2}^{\infty} \frac{\lambda^{k-2} e^{-\lambda}}{(k-2)!} + \lambda = \lambda^2 \sum_{k=0}^{\infty} \mathbb{P}(X = k) + \lambda \\ &= \lambda^2 + \lambda.\end{aligned}$$

In the above equalities, we used $\frac{k^2-k}{k!} = \frac{k(k-1)}{1 \cdot 2 \dots (k-1) \cdot k} = \frac{1}{(k-2)!}$. Hence

$$\text{Var}(X) = \lambda + \lambda^2 - \lambda^2 = \lambda.$$

Exercise 3 (iPhone Headphone Jacks)

Three different factories in Asia are capable of making the new iPhone. The factories' names are A , B , and C and they are all of different quality. If factory A is given the newest iPhone contract they will accidentally add headphone jacks on a number of phones. For the duration of one week, this number has a Poisson distribution with mean $a = 100$. If factory B does the job the parameter is $b = 200$ and if factory C does the job the parameter is $c = 300$. Suppose that each factory is equally likely to be chosen.

- Let X be the number of iPhones that receive headphone jacks throughout the year. Suppose each week Apple places an order independently across the three factories. Compute the mean and variance of X .
- By the end of the first week no iPhones are found to be defective. What is the likelihood that Factory A received the contract?

Solution:

- Let X_i be the number of defective iPhones in week i and Y_i be the factory receiving the contract that week. Thus, conditioning on $Y_i = A$, X_i follows a Poisson distribution with mean $a = 100$, which means that

$$\mathbb{E}(X_i|Y_i = A) = 100, \quad \text{Var}(X_i|Y_i = A) = 100,$$

and thus $\mathbb{E}(X_i^2|Y_i = A) = \text{Var}(X_i|Y_i = A) + [\mathbb{E}(X_i|Y_i = A)]^2 = 10100$. For the same reason

$$\mathbb{E}(X_i|Y_i = B) = 200, \quad \mathbb{E}(X_i^2|Y_i = B) = 40,200$$

$$\mathbb{E}(X_i|Y_i = C) = 300, \quad \mathbb{E}(X_i^2|Y_i = C) = 90,300$$

Note that the problem is asking us to calculate the number of defective iPhones during the year, which in our new notation will be $X = \sum_{i=1}^{52} X_i$. According to the law of total expectation,

$$\begin{aligned} \mathbb{E}[X_i] &= \mathbb{E}[\mathbb{E}(X_i|Y_i)] \\ &= \mathbb{P}(Y_i = A)\mathbb{E}(X_i|Y_i = A) + \mathbb{P}(Y_i = B)\mathbb{E}(X_i|Y_i = B) + \mathbb{P}(Y_i = C)\mathbb{E}(X_i|Y_i = C) \\ &= 100/3 + 200/3 + 300/3 = 200. \end{aligned}$$

Moreover,

$$\begin{aligned} \mathbb{E}[X_i^2] &= \mathbb{E}[\mathbb{E}(X_i^2|Y_i)] \\ &= \mathbb{P}(Y_i = A)\mathbb{E}(X_i^2|Y_i = A) + \mathbb{P}(Y_i = B)\mathbb{E}(X_i^2|Y_i = B) + \mathbb{P}(Y_i = C)\mathbb{E}(X_i^2|Y_i = C) \\ &= 10,100/3 + 40,200/3 + 90,300/3 = 140,600/3. \end{aligned}$$

Therefore,

$$\text{Var}(X_i) = \mathbb{E}[X_i^2] - (\mathbb{E}X_i)^2 = 20,600/3.$$

Now, we are ready to answer the original question, by the linearity of expectation we get:

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^{52} X_i\right] = \sum_{i=1}^{52} \mathbb{E}[X_i] = 52\mathbb{E}[X_1] = 10,400.$$

Since the choice of the factory and the production of iPhones is independent week to week, we know that the random variables X_i are independent and identically distributed. Thus, we can easily calculate the variance of X :

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^{52} X_i\right) = \sum_{i=1}^{52} \text{Var}(X_i) = 52 \text{Var}(X_1) \approx 357,066.$$

- (b) Following the analysis for the previous question, noticing the fact that conditional on $Y = A$, X follows a Poisson distribution with mean $a = 100$ (in this question we consider only one week), we have that

$$\mathbb{P}(X = 0|Y = A) = e^{-a} = e^{-100}.$$

Similarly

$$\mathbb{P}(X = 0|Y = B) = e^{-200}, \quad \mathbb{P}(X = 0|Y = C) = e^{-300}.$$

According to Bayes formula,

$$\begin{aligned} & \mathbb{P}(Y = A|X = 0) \\ &= \frac{\mathbb{P}(Y = A)\mathbb{P}(X = 0|Y = A)}{\mathbb{P}(Y = A)\mathbb{P}(X = 0|Y = A) + \mathbb{P}(Y = B)\mathbb{P}(X = 0|Y = B) + \mathbb{P}(Y = C)\mathbb{P}(X = 0|Y = C)} \\ &= \frac{1/3 \cdot \exp(-100)}{1/3 \cdot \exp(-100) + 1/3 \cdot \exp(-200) + 1/3 \cdot \exp(-300)} \\ &\approx 1. \end{aligned}$$

Exercise 4 (Returning Exams)

Students of a class whose concentration and course code must remain anonymous have written their midterm exam and are anxiously awaiting for their results. Finally, an e-mail arrives telling them to pick up their exams from their individual folders. Unfortunately, the AI who was supposed to distribute their exams was running on close to zero sleep and ended up placing the exams in a completely random order. If the class has 140 students, how many of them on average are expected to get their exams, and what is the variance of this number?

Solution:

Consider a random permutation of the set $\{1, 2, \dots, n\}$, and let X be the number of fixed points (numbers which are in the same spot before and after the permutation). Our posed problem is equivalent to evaluating $E(X)$ and $Var(X)$. Let X_i be the indicator that i is a fixed point, and note that $X = X_1 + \dots + X_n$. The indicator random variables are *not* independent in this case! Let us first calculate the expectation:

$$\mathbb{E}\left[\sum_{i=1}^n X_i\right] = n\mathbb{E}[X_i] = n \times \frac{1}{n} = 1.$$

Now, let's calculate the variance! Note X_1X_2 is also an indicator random variable, so $\mathbb{E}[X_1X_2]$ is the probability that both 1 and 2 are fixed points:

$$\begin{aligned}
\text{Var}(X) &= \text{Var}\left(\sum_{i=1}^n X_i\right) = \mathbb{E}\left[\left(\sum_{i=1}^n (X_i - \mathbb{E}[X_i])\right)^2\right] \\
&= \sum_{i=1}^n \mathbb{E}[(X_i - \mathbb{E}[X_i])^2] + \sum_{i \neq j} \mathbb{E}[(X_i - \mathbb{E}[X_i])(X_j - \mathbb{E}[X_j])] \\
&= n \text{Var}(X_1) + n(n-1) \mathbb{E}[(X_1 - \mathbb{E}[X_1])(X_2 - \mathbb{E}[X_2])] \\
&= n \cdot \frac{1}{n} \left(1 - \frac{1}{n}\right) + n(n-1) \cdot (\mathbb{E}[X_1X_2] - \mathbb{E}[X_1]\mathbb{E}[X_2]) \\
&= \frac{n-1}{n} + n(n-1) \cdot \left(\frac{1}{n(n-1)} - \frac{1}{n^2}\right) \\
&= \frac{n-1}{n} + \frac{1}{n} \\
&= 1.
\end{aligned}$$